

Ectopic pregnancy

➤ Objectives

- ✓ By the end of this lesson you should be able to:
 - Define ectopic pregnancy
 - Describe clinical manifestations of EP
 - Describe the sites of ectopic pregnancy
 - Explain the causes/risk factors for ectopic pregnancy
 - Discuss about how to diagnose EP
 - Explain the complications and managements of EP

➤ Definition

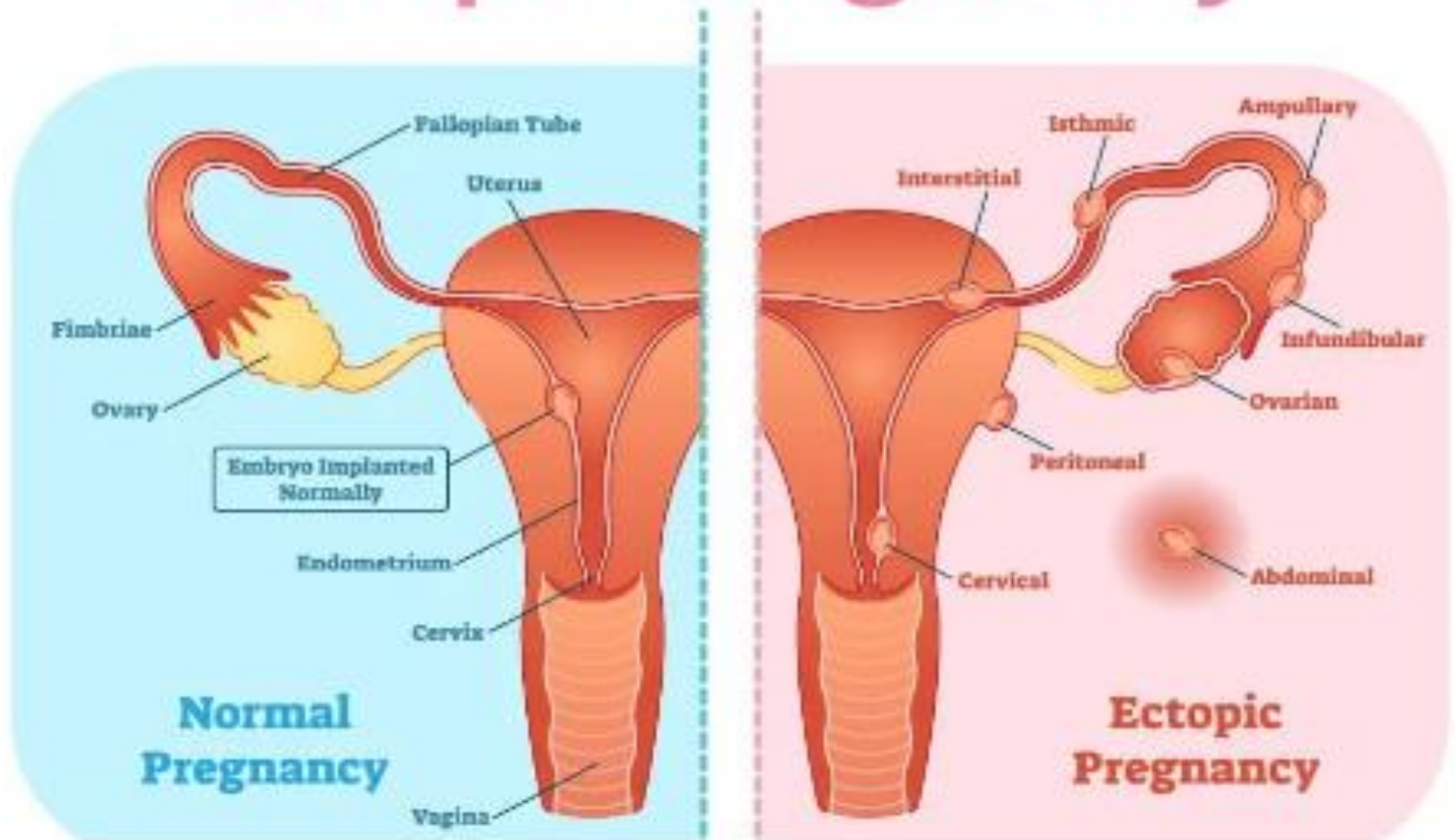
- An ectopic pregnancy is implantation of a fertilized ovum implants anywhere other than the endometrial lining of the uterine cavity
- It is also called "**an out-of-place pregnancy**"
- the fallopian tube is the most common site of implantation which accounts for 98 percent of all ectopic gestations

Classification

- ❖ **Tubal pregnancy** (99%)
 - Ampullary (mid) (80-90%)
 - Isthmic (5-10%)
 - Fimbrial (5%)
 - Interstitial (cornual) (1-2%)
- ❖ **Abdominal** (1-2%) primary/secondary (tubo-abdominal/abdomino-ovarian)
- ❖ **Ovarian** (0.5-1%)
- ❖ **Cervical** (less than 0.5%)
- ❖ **Heterotopic** (combination of ectopic + intrauterine pregnancy)

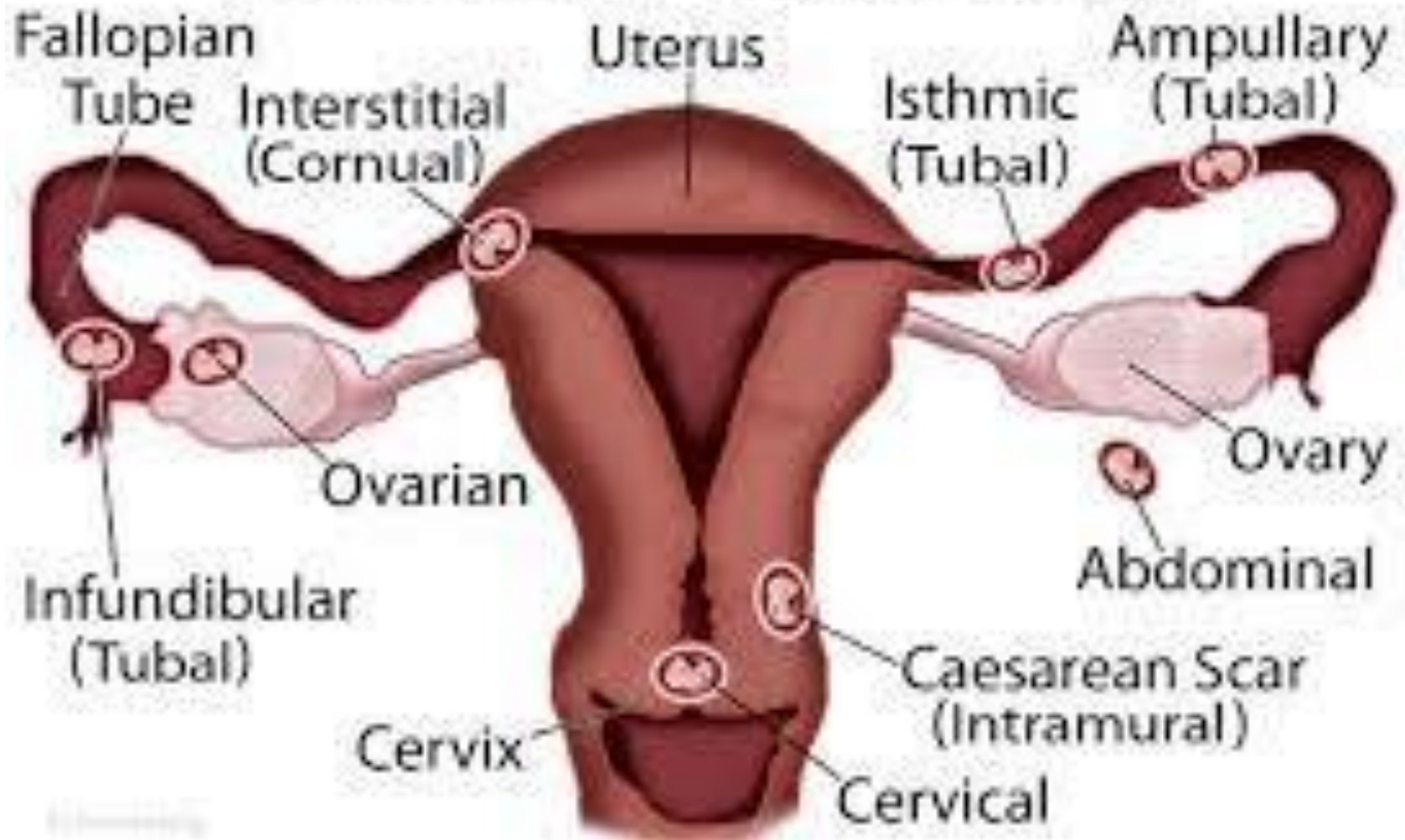
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Ectopic Pregnancy



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Ectopic Pregnancy



Risk factors

➤ High Risk

- Tubal surgery
- Tubal ligation
- Previous ectopic pregnancy
- In utero exposure to DES
- Use of IUD
- Tubal pathology
- Assisted reproduction

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➤ Moderate risk

- Infertility
- Previous genital infections
- Multiple sexual partners
- Salphingitis

➤ Low risk

- Previous pelvic infection
- Cigarette smoking
- Vaginal Douching
- First intercourse less than 18 years

Diagnosis of ectopic pregnancy

➤ **History** — The classic triad symptoms of ectopic pregnancy are:

- Abdominal pain
- Amenorrhea
- Vaginal bleeding

✓ Blood leaking out of the fallopian tube may irritate the diaphragm and cause **shoulder pain**, whereas blood pooling in the posterior cul-de-sac (pouch of Douglas) may cause an **urge to defecate**

✓ **Lightheadedness or shock** suggests tubal rupture has occurred

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➤ Physical examination

- Clinical presentations are extremely variable depending on whether the tube has ruptured or not

Sign	Unruptured	Ruptured
Vital Signs	• Vital signs can be normal or slightly deranged	• Fast, weak pulse • Hypotension • Pallor
Abdominal examination	• Non-tender abdomen	• Direct abdominal tenderness • Rebound tenderness • Abdominal distension • Bowel sounds are decreased
Pelvic examination	• Minimal vaginal bleeding • Adnexal mass	• Minimal vaginal bleeding • Cervical motion tenderness • Adnexal tenderness

Cont...

➤ Investigations:

- **Ultrasound**
- **HCG**
- **Diagnostic laparoscopy : Gold standard** for diagnosis of ectopic pregnancy
- **MRI**
- **Culdocentesis**
- **Progesterone level determination:**
 - progesterone levels are higher in intrauterine pregnancies than in ectopic pregnancies

Differential diagnosis

- UTI
- kidney stones
- appendicitis
- ovarian neoplasms
- endometriosis
- endometritis
- Leiomyomas
- PID
- Threatened Abortion
- Ruptured ovarian cyst
- Salpingitis
- Ovarian torsion & Intrauterine pregnancy

Natural history ectopic pregnancy

- ✓ If left untreated, an ectopic pregnancy in the fallopian tube can progress to;
 - tubal abortion
 - tubal rupture, or
 - it may regress spontaneously

Management Options of ectopic pregnancy

✓ In all cases, ectopic pregnancy must be terminated

1. Medical Management : Oral, parenteral or direct Injection into Ectopic Pregnancy

▪ Methotrexate

❖ Indications for Methotrexate

- Stable vital signs with normal LFTs, CBC, platelets
- Hemodynamically stable
- Unruptured ectopic pregnancy without cardiac activity
- Small ectopic pregnancy size (less than 3.5 cm mass)
- β HCG <5000 IU/L
- Willing and able to comply to post treatment follow-up

Cont...

❖ Contraindications for Methotrexate (anti-metabolite)

- Hemodynamically unstable
- hepatic/renal dysfunction
- Active pulmonary disease
- Immunodeficiency,
- peptic ulcer disease
- Hypersensitivity to Methotrexate
- Breast feeding
- Tubal rupture
- Presence of intrauterine pregnancy (heterotopic pregnancy)

- unable to be compliant with post-therapeutic monitoring

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❖ Administration of Methotrexate

- Methotrexate maybe administered in a single intramuscular dose or in multiple doses
- ✓ Single-dose methotrexate protocol for ectopic pregnancy
 - Methotrexate 50 mg/m² IM on day 1
 - Determine serum B-hCG level on day 1, day 4 and day 7
 - Once 15% decline in β -hCG level is achieved between day 4 and day 7, serum β -hCG level weekly until undetectable (average time for resolution is 36 days)
 - If decline of 15% is not achieved repeat Methotrexate 50 mg/m² IM on day 7 or resort to surgical management.

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2. Expectant Management Indications

- Minimal pain or bleeding in reliable patient
- Beta HCG less than 1000 IU/L and falling
- No signs of tubal rupture
- Adnexal mass <3 cm
- Tubal location

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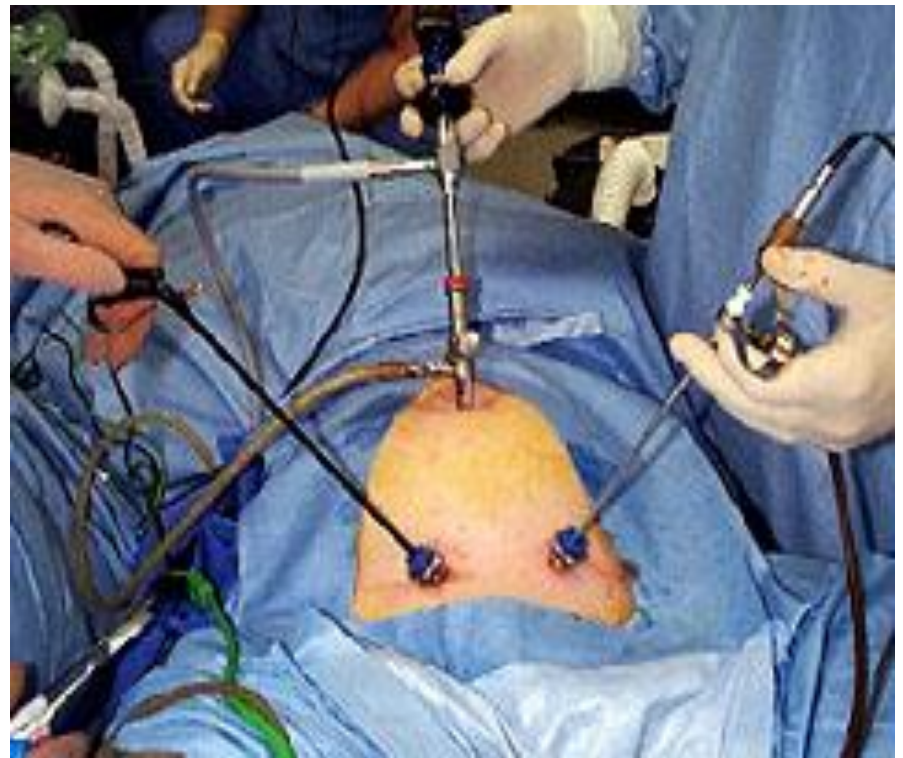
3. Surgical -Laparotomy & Laparoscopic

- Salpingectomy
- Salpingotomy/Salpingostomy (conservative surgery)
- ❖ **Specific indications for surgical therapy include:**
 - Hemodynamic instability
 - Impending or ongoing ectopic mass rupture
 - Not able or willing to comply with medical therapy post treatment follow-up
 - Lack of timely access to a medical institution for management of tubal rupture
 - Failed medical therapy
 - Hemoperitoneum
 - Non diagnostic Transvaginal US and β HCG >1500
 - Advanced ectopic pregnancy

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1. Laparoscopy

- Recommended approach
- Usually, if the tube is not ruptured



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2. Laparotomy

- Surgical incision through the abdominal wall
- Pfannensteil incision
- Mainly used for cases involving haemodynamic instability



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❖ Subsequent Management

- Correct anemia
- Rh-immunoglobulin (Anti-D) should be administered to all Rh-negatives
- Provide family planning counseling and service Provide counseling and advice on prognosis of fertility before discharge
- Given the increased risk of a future ectopic pregnancy, patients should also be counseled to come as soon as she misses her menses
- Schedule a follow-up visit

Post-therapy monitoring and evaluation

- Serum beta-hCG concentration and ultrasound examination should be evaluated weekly
- An increase in beta hCG levels in the three days following therapy (i.e., up to day 4) and mild abdominal pain of short duration (one to two days) are common
- The pain can be controlled with nonsteroidal anti-inflammatory drugs

Gestational trophoblastic disease (GTD)

- GTD is a proliferative disorder of trophoblastic cells arising from the trophoblastic epithelium of the placenta
- Is a broad spectrum of **benign and malignant** tumors derived from the trophoblastic of the human placenta
- All forms of GTD are characterized by a distinct tumor marker, the beta subunit of human chorionic gonadotropin (hCG)
- The pathogenesis of GTD is unique because *the maternal tumor arises from gestational rather than maternal tissue.*

Types of GTD

- Hydatidiform mole (complete or partial)
- Persistent/invasive gestational trophoblastic neoplasia (GTN)
- Choriocarcinoma
- Placental site trophoblastic tumors

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- Hydatidiform moles are noninvasive, localized/**benign** tumors that develop as a result of an aberrant fertilization event that leads to a proliferative process
- They comprise 90 percent of GTD cases
- The other **three** categories of GTD represent **malignant** disease because of their potential for local invasion and metastases.

Risk factors of GTD

- The two main risk factors for GTD are extremes of maternal age (over age 35 years) and a history of previous GTD
- **Extremes of age** — the most well-established risk factor for GTD is maternal age
- The risk is significantly increased in those older than age 35 and slightly increased in those under age 20
- Paternal age does not appear to influence the risk of GTD

Clinical manifestations of GTD

✓ Clinical manifestations of GTD include, in decreasing order of frequency

- Vaginal bleeding/AUB-90%
- Enlarged uterus
- Pelvic pressure or pain
- Theca lutein cysts
- Anemia
- Hyperemesis gravidarum
- Hyperthyroidism (HCG mediated)
- Preeclampsia before 20 weeks of gestation
- Vaginal passage of hydropic vesicles
- Clinical hyperthyroidism develops in approximately 5 % of women with GTD

Hydatidiform mole (molar pregnancy)

- Molar pregnancy is characterized histologically by abnormalities of the chorionic villi that consist of trophoblastic proliferation and edema of villous stroma
- Complete and partial Hydatidiform moles are noninvasive, localized/**benign** tumors that develop as a result of an aberrant fertilization event that leads to a proliferative process

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- Women with this disorder typically present to their obstetrical provider with missed menstrual periods, a positive pregnancy test, and signs and symptoms consistent with early pregnancy.
- The absence or presence of a fetus or embryonic elements has been used to describe them as *complete and partial moles*.

Characteristics of complete & partial Hydatidiform mole

❖ Complete Hydatidiform mole

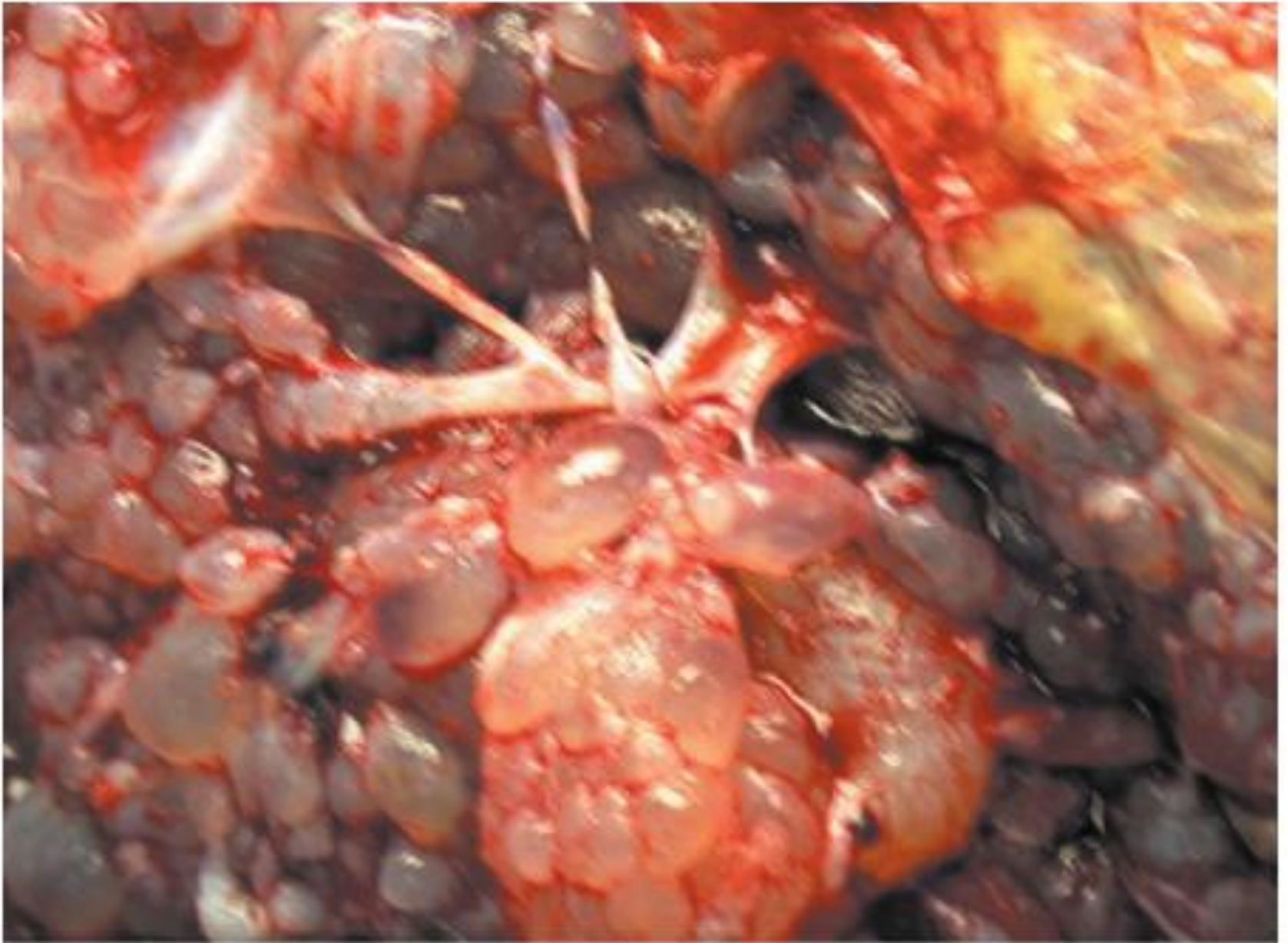
✓ A complete mole is characterized by;

- fertilization of an empty ovum by two sperms or a single sperm that duplicates, resulting in a 46 XX or 46 XY karyotype
- lacks a fetus and often leads to excessive uterine size for the expected "gestational age"
- Uterine enlargement is due to the tumor itself and/or intrauterine hemorrhage with retained clot
- associated with marked elevation in serum hCG

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✓ Sonographic features of Complete Hydatidiform mole :

- Absence of an embryo or fetus
- No amniotic fluid
- a "snowstorm pattern" on older ultrasounds
- Theca lutein cysts



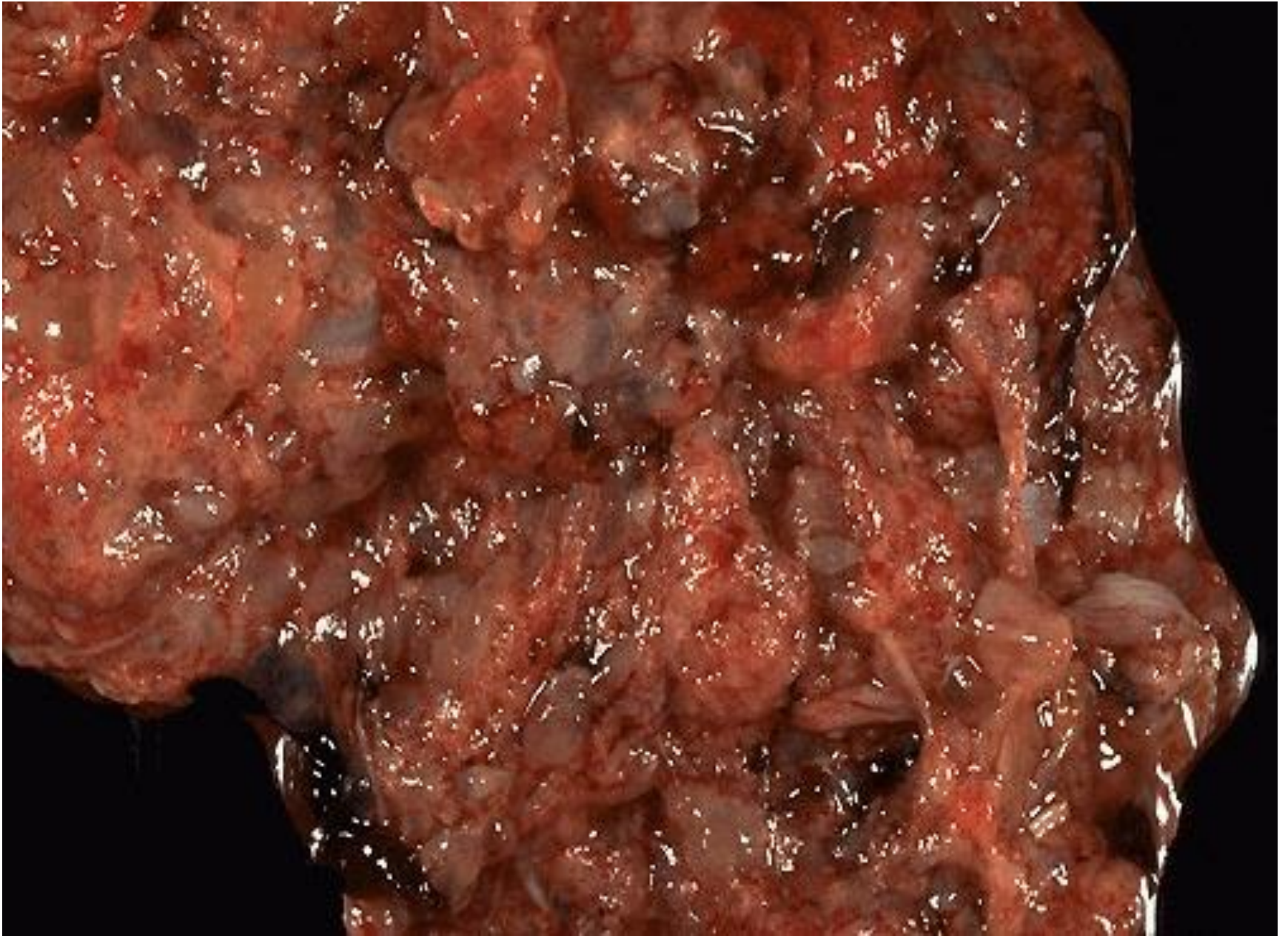
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❖ Partial Hydatidiform mole

- ✓ A partial mole is characterized by;
 - fertilization of a haploid ovum by two sperm or duplication of one sperm,
 - Triploid karyotype (69 XXY, 69 XXX, 69 XYY)
 - Partial moles are the only type of GTD that are associated with *the presence of a fetus, and fetal cardiac activity* may be detected.

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- ✓ These pregnancies are infrequently associated with;
 - excessive uterine size
 - ovarian enlargement
 - preeclampsia, hyperemesis, or hyperthyroidism because hCG levels are generally lower than those observed with a complete mole.



Diagnosis of Hydatidiform mole

- When complete or partial mole is suspected based upon sonographic findings and hCG determination, the diagnosis must be confirmed by **histologic examination of tissue**
- The routine availability of both ultrasonography and sensitive quantitative measurement of the serum concentration of the beta subunit of hCG has led to earlier diagnosis of GTD, often before complications are manifested

Mangement

Surgery

Initial management of suspected complete or partial mole is **evacuation of the uterine contents by suction curettage.**

- This procedure is a definitive therapy for most patients.
- Patients who have no desire for future fertility may opt for hysterectomy, which eliminates the risk of local invasion.
- **Preoperative testing** — A baseline hCG level should be obtained for comparison with postoperative results.

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- A complete blood count, and renal, liver, and thyroid function tests can be used.
- blood should be available and adequate venous access assured.
- We generally perform these procedures under general anesthesia.
- Under adequate anesthesia, oxytocin is administered and the cervix is dilated to allow passage of the suction cannula.

POST MOLAR FOLLOW UP

- Following molar evacuation, patients must be monitored for development of post molar GTN
 - 18 to 28 % of patients with complete mole develop persistent disease.
 - But the risk of GTN after partial molar pregnancy is no more than 2 to 4 percent
- prophylactic chemotherapy are given at the time of molar evacuation reduces the risk of post molar GTN

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❖ HCG level : after molar evacuation

1) **Monitored weekly until normal for 3 wk**

2) **Follow monthly until normal for 6 month**

- The average time to achieve first normal hCG level after evacuation is about 9 wk

3) **Contraception**

THANK YOU